

Case Study Practical

Task 1: Laptop Won't Power On (5 Troubleshooting Steps)

1. Verify the AC Power Outlet & Charger LED:

- *What you are checking for:* Test the wall socket with another working device (like a phone charger) and check if the laptop charger's indicator light is glowing to verify incoming electrical power.

2. Inspect the Charging Cable & DC Jack:

- *What you are checking for:* Look for cuts, frayed wires, or bent pins inside the charger plug and laptop charging port, and check if the connector feels excessively loose or damaged.

3. Check the Laptop Charging Status LED:

- *What you are checking for:* See if the laptop's battery/power indicator LED illuminates or blinks when plugged in to determine if the motherboard is receiving standby voltage.

4. Perform a Hard Power Reset (Drain Residual Static):

- *What you are checking for:* Disconnect the charger, remove the battery (if external/accessible), and press and hold the power button for 30 seconds. This drains trapped static charge from the motherboard capacitors that can lock the power circuit.

5. Test with Battery Disconnected (or Alternate Charger):

- *What you are checking for:* Power the laptop using only the AC adapter without the battery installed to identify whether a shorted/dead battery is preventing the system from turning on.

Task 2: Helpdesk Ticket: 'Monitor Blank Screen, CPU Running' (Step-by-Step Flow)

- **Step 1: Check Monitor Power & Status LED**
 - Verify the monitor's power cord is firmly plugged in and the power button is on. Observe the power LED:
 - *Amber / Blinking*: Monitor is receiving power but no video signal.
 - *Unlit*: Monitor has no power.
- **Step 2: Verify Video Cable & Port Location**
 - Check that the HDMI/DisplayPort cable is securely seated on both ends. Ensure the cable is plugged into the **dedicated graphics card (GPU)** at the bottom rear of the PC, not the motherboard video port.
- **Step 3: Check Monitor Input Source & Brightness Settings**
 - Use the monitor's physical menu buttons (OSD) to confirm the input source matches the cable used (e.g., set to *HDMI 1* or *DP 1* instead of *Auto-detect*).
- **Step 4: Wake Display via Keyboard Shortcut**
 - Press **Win + Ctrl + Shift + B** on the keyboard. This restarts the Windows graphics driver; a brief beep confirms the OS is active and responsive.
- **Step 5: Test with Known-Good Cable / Alternate Display**
 - Swap the HDMI/DP cable with a known-working one, or connect the PC to an alternate monitor/TV to isolate whether the display or cable is defective.
- **Step 6: Internal Hardware Inspection (Reseat RAM & GPU)**
 - If the screen remains blank on all displays, power down, open the PC, and reseat the RAM sticks and GPU in their motherboard slots.

Task 3: Wireless Mouse Unresponsive (Isolating Mouse, USB Port, or Settings)

- **1. Isolate the Mouse (Hardware & Power):**
 - Turn the power switch on the bottom of the mouse **ON**.
 - Replace the AA/AAA battery with a fresh one (or recharge it) and verify the optical LED or sensor light turns on.
 - Connect the mouse to another working computer; if it still does not work, the mouse hardware or internal wireless transmitter is defective.
- **2. Isolate the USB Port:**
 - Unplug the 2.4 GHz USB wireless receiver (dongle) and plug it into a different USB port (preferably directly into the rear motherboard port, avoiding unpowered hubs).
 - Plug a known-working USB flash drive into the original port to verify if that specific USB port provides power and data connectivity.
- **3. Isolate the System Settings & Drivers:**
 - Press **Win + X** and open **Device Manager** using the keyboard.
 - Check under "**Mice and other pointing devices**" and "**Universal Serial Bus controllers**" for yellow warning triangles (!), disabled devices, or driver errors.
 - Uninstall the mouse driver, unplug and re-insert the USB receiver, and allow Windows to automatically reinstall the driver.

Task 4: Gaming Café System Overheating & Shutting Down (3 Hardware Causes)

1. Dust Clogging the Heatsink & Chassis Airflow:

- *Cause:* Heavy dust and lint build up inside the CPU/GPU heatsink fins and mesh intake filters, blocking heat from dissipating.
- *How to check:* Open the side case panel and visually inspect the heatsink fin stacks and dust filters under a flashlight. Clean using compressed air or an electric duster.

2. Cooling Fan Failure or Low RPM:

- *Cause:* A seized motor, disconnected fan header, or broken bearing prevents the CPU cooler fan or chassis intake/exhaust fans from spinning at required speeds.
- *How to check:* Turn on the PC and visually inspect if all fans spin smoothly without rattling, or run hardware monitoring software (e.g., HWMonitor / BIOS Hardware Monitor) to verify the CPU fan is reporting normal RPM (>1000 RPM under load).

3. Dried / Degraded Thermal Paste:

- *Cause:* High-performance gaming dries out and cracks the thermal compound between the CPU die and cooler baseplate over time, creating microscopic air pockets that trap heat.
- *How to check:* Remove the CPU cooler heatsink and examine the thermal paste on the processor. If it is dry, flaky, or powdery rather than smooth and viscous, clean it off using 99% isopropyl alcohol and apply fresh thermal compound before remounting.